

Dynamic Macroeconomics II

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Fall 2019

Optimal Fiscal Policy with Commitment

First Part

Version reviewed September 9 2019

- We formulate the dynamic optimal taxation problem called the **Ramsey Problem**.
- Which taxes maximize household's welfare?
- Solution to this problem is called **Ramsey plan**.
- Government can issue public debt.
- No uncertainty.
- The government **commits** to future tax rates:
This means that it will not deviate in the future from tax plan announced today.
- Main result: capital should **not** be taxed in **long run**.
- This is result for versions of neoclassical growth model.
Not true in general.

- Utility over consumption and leisure:

$$\sum_{t=0}^{\infty} \beta^t u(c_t, 1 - n_t) \quad 0 < \beta < 1$$

- u is strictly increasing in c_t and $1 - n_t$, twice continuously differentiable and strictly concave.
- u_{ct} and u_{nt} denote marginal utilities of consumption and labor, respectively; $u_{leisure t} = u_{lt}$.
- Technology $F(k_t, n_t)$: homogeneous of degree one with positive and diminishing marginal returns in both inputs.
- By Euler's theorem: $F(k_t, n_t) = F_k(t)k_t + F_n(t)n_t$.
- Capital accumulation: $k_{t+1} = (1 - \delta)k_t + x_t$, where x_t is gross investment.
- Feasibility condition: $y_t = c_t + x_t + g_t$.

- Imposes sequences of taxes $\{\tau_{k_t}, \tau_{n_t}\}_{t=0}^{\infty}$ on capital and labor earnings.
- Purchases exogenous stream $\{g_t\}_{t=0}^{\infty}$.
We keep this stream unchanged, will look for optimal taxes to finance it.
- Trades one-period bonds b_t , with gross rate of return R_t .
- Government's sequential budget constraint is given by:

$$g_t = \tau_{k_t} r_t k_t + \tau_{n_t} w_t n_t + \frac{b_{t+1}}{R_t} - b_t.$$

- Maximize utility subject to budget constraint:

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$$c_t + k_{t+1} - (1 - \delta)k_t + \frac{b_{t+1}}{R_t} - b_t = (1 - \tau_{nt})w_t n_t + (1 - \tau_{kt})r_t k_t.$$

- First order conditions for consumer's optimization problem, with Lagrange multiplier $\beta^t \lambda_t$ (irrelevant for system of equations below, **check**):

$$c_t : u_c(t) = \lambda_t$$

$$n_t : u_l(t) = \lambda_t(1 - \tau_{nt})w_t$$

$$k_{t+1} : \lambda_t = \beta \lambda_{t+1} [(1 - \tau_{kt+1})r_{t+1} + 1 - \delta]$$

$$b_{t+1} : \lambda_t \frac{1}{R_t} = \beta \lambda_{t+1}$$

- Euler equation:

$$u_{ct} = \beta u_{ct+1} [(1 - \tau_{kt+1})r_{t+1} + (1 - \delta)]$$

- Consumption-leisure equation:

$$u_{lt} = u_{ct}(1 - \tau_{nt})w_t$$

- No-arbitrage condition:

$$R_t = (1 - \tau_{kt})r_{t+1} + (1 - \delta)$$

- No-arbitrage condition can be obtained, **as we have seen before**, by consolidating two consecutive budget constraints and eliminating common term b_{t+1} :

$$c_t + \frac{c_{t+1}}{R_t} + \frac{k_{t+2}}{R_t} + \frac{b_{t+2}}{R_t R_{t+1}} = b_t + (1 - \tau_{nt})w_t n_t + \frac{(1 - \tau_{nt+1})w_{t+1}n_{t+1}}{R_t} + (1 - \tau_{kt})r_t k_t + (1 - \delta)k_t + \left[\frac{(1 - \tau_{kt+1})r_{t+1} + 1 - \delta}{R_t} - 1 \right] k_{t+1}$$

- To ensure the existence of an equilibrium

$$\frac{(1 - \tau_{kt+1})r_{t+1} + 1 - \delta}{R_t} - 1 = 0$$

must hold $\forall t \geq 0$.

- Repeat the previous operation to get household's present-value budget constraint.
- In this model interest rate is **not** constant. Therefore

$$\sum_{t=0}^{\infty} \left(\prod_{i=0}^{t-1} R_i^{-1} \right) c_t = \sum_{t=0}^{\infty} \left(\prod_{i=0}^{t-1} R_i^{-1} \right) (1 - \tau_{nt}) w_t n_t + [(1 - \tau_{k0}) r_0 + 1 - \delta] k_0 + b_0.$$

(minor issue with discount factor at $t=0$)

- Transversality conditions:

$$\lim_{T \rightarrow \infty} \left(\prod_{i=0}^{T-1} R_i^{-1} \right) k_{T+1} = 0$$

$$\lim_{T \rightarrow \infty} \left(\prod_{i=0}^{T-1} R_i^{-1} \right) \frac{b_{T+1}}{R_T} = 0 \quad \left\{ \begin{array}{l} \text{have} \\ \text{upper case} \end{array} \right.$$

- Take prices (r_t, w_t) as given and maximize profits:

$$\Pi_t = F(k_t, n_t) - r_t k_t - w_t n_t$$

- First order conditions:

$$F_k(t) = r_t$$

$$F_n(t) = w_t$$

- In any equilibrium, these conditions imply that $\Pi = 0$.

The Ramsey Problem

- Symbol without subscript denote infinite sequence for corresponding variable: e.g. $c \equiv \{c_t\}_{t=0}^{\infty}$.
- **Feasible allocation**: sequence (k, c, ℓ, g) that satisfies feasibility.
- **Price system**: nonnegative bounded sequences (w, r, R) .
- **Government policy**: sequence (g, τ_k, τ_n, b) .

- **Competitive equilibrium:** a feasible allocation, a price system and a government policy such that:
 - 1 Given price system and government policy, feasible allocation solves both household's problem and firm's problem.
 - 2 Given feasible allocation and price system, government policy satisfies government's budget constraint.
- There are many competitive equilibria, indexed by different government policies.
- Which one is the best equilibrium from the point of view of the consumer? I.e. which government policy maximizes welfare measured with utility function?

- Given k_0 and b_0 , **Ramsey Problem** is to choose a competitive equilibrium that maximizes household's utility.
- To make problem interesting, we **impose** restriction on τ_{k_0} : it must be a small number, say zero.
- By doing so, avoid possible simple solution to Ramsey Problem:
tax only k_0 , which causes no distortion anything since it is as given (inelastic).

Chamley's approach to Ramsey Problem

- Following Chamley (1986), formulate Ramsey Problem as if government chooses after-tax prices of inputs:

$$\begin{aligned}\bar{r}_t &= (1 - \tau_{kt})r_t \\ \bar{w}_t &= (1 - \tau_{nt})w_t\end{aligned}$$

- Express government's tax revenues as:

$$\begin{aligned}\tau_{kt}r_tk_t + \tau_{nt}w_tn_t &= (r_t - \bar{r}_t)k_t + (w_t - \bar{w}_t)n_t \\ &= F_k(t)k_t + F_n(t)n_t - \bar{r}_tk_t - \bar{w}_tn_t \\ &= F(k_t, n_t) - \bar{r}_tk_t - \bar{w}_tn_t\end{aligned}$$

Express government's budget constraint:

$$g_t = F(k_t, n_t) - \bar{r}_tk_t - \bar{w}_tn_t + \frac{b_{t+1}}{R_t} - b_t$$

- Allocation that solves Ramsey Problem must be a competitive equilibrium. Form a Lagrangian: utility is constrained by government constraint, feasibility, and by **consumer's choices**

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$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \left\{ U(c_t, 1 - n_t) + \right. \\ \left. \psi_t \left[F(k_t, n_t) - \bar{r}_t k_t - \bar{w}_t n_t + \frac{b_{t+1}}{R_t} - b_t - g_t \right] + \right. \\ \left. \theta_t [F(k_t, n_t) - c_t - k_{t+1} + (1 - \delta)k_t - g_t] \right. \\ \left. + \mu_{1t} [u_l(t) - u_c(t)\bar{w}_t] \right. \\ \left. + \mu_{2t} [u_c(t) - \beta u_c(t+1)(\bar{r}_{t+1} + 1 - \delta)] \right\}$$

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where $R_t = \bar{r}_{t+1} + 1 - \delta$ (No Arbitrage).

- Consumer's budget not a constraint, implied by govt. constraint and feasibility: **Walras Law**. [Se eu vaciar n-1 mkt, ent n fmb se vacia]
- First order condition with respect to k_{t+1} (**check**):

$$\theta_t = \beta \{ \psi_{t+1} [F_{k_{t+1}} - \bar{r}_{t+1}] + \theta_{t+1} [F_{k_{t+1}} + 1 - \delta] \}$$

Como cambia la revalorización si mañana hay un poco más de capital físico

Interpretation:

- Marginal increment in capital investment in t produces $F_k(t+1) + 1 - \delta$ at time $t+1$
- It also produces a marginal increase in govt. revenue of $F_k(t+1) - \bar{r}_{t+1}$, which allows govt. to reduce debt or some tax.
- Note that θ_t and ψ_t are **positive** Lagrange multipliers that represent the marginal value of an extra marginal unit of production (θ_t), and the marginal value of an extra marginal unit of govt. revenue. (ψ_t)

Calculation of steady-state optimal tax on capital

- Suppose government expenditures g_t remain constant after some period T .
- Assume solution of Ramsey Problem converges to a steady state.
- Steady-state version of first-order condition for capital:

$$\theta = \beta \{ \psi [F_k - \bar{r}] + \theta [F_k + 1 - \delta] \}$$

- From firm's problem, substitute $r_t = F_{kt}$ to yield:

$$\theta = \beta \{ \psi [r - \bar{r}] + \theta [r + 1 - \delta] \}$$

- Keep in mind the steady-state Euler equation **of the consumer**:

$$1 = \beta(\bar{r} + 1 - \delta)$$

- Combine the previous two equations to get (**check**):

$$(\theta + \psi)(r - \bar{r}) = 0$$

- θ is strictly positive, as utility is strictly increasing, therefore production will be exhausted.
- ψ is nonnegative, which we said before.
- Therefore, $\bar{r} = r$, which implies $\tau_k = 0$.
- Conclusion: if there exists steady-state allocation that solves Ramsey Problem, associated tax rate on capital earnings must be zero.
- Result robust to eliminating govt. debt (set $b_{t+1} = b_t = 0$ in Lagrangian): does not affect our derivation.

- Explores limits to redistribution.
- Or: does previous result hold if Ramsey cares only about workers who do not own capital?
- $i = 1, \dots, N$ agents with different utilities $u_i(c_{it}, 1 - n_{it})$.
- All agents with same discount factor (below: what would happen otherwise?).
- Lump-sum transfers: $S_{it} \geq 0$.
- Government has a social welfare function that is a weighted average of individual utilities with weight $\alpha_i \geq 0$ on individual i :

$$\sum_{i=1}^N \alpha_i u_i(c_{it}, 1 - n_{it}), \quad \alpha_i \geq 0, \quad \sum_{i=1}^N \alpha_i = 1.$$

- Government has balanced budget (can change that, result the same).

- Household's budget constraint:

$$c_{it} + k_{it+1} - (1 - \delta)k_{it} = (1 - \tau_{nt})w_t n_{it} + (1 - \tau_{kt})r_t k_{it} + S_{it}$$

All households face same prices and taxes.

- Household's budget constraint (with net taxes as Chamley):

$$c_{it} + k_{it+1} - (1 - \delta)k_{it} = \bar{w}_t n_{it} + \bar{r}_t k_{it} + S_{it}$$

- Below we are going to use notation

$$x_t \equiv \sum_{i=1}^N x_{it}, x = c, n, k, S$$

- Government's budget constraint:

$$g_t + S_t = \tau_{kt} r_t k_t + \tau_{nt} w_t n_t.$$

- Government's budget constraint (with net taxes):

$$g_t + S_t = F(k_t, n_t) - \bar{r}_t k_t - \bar{w}_t n_t.$$



- We now construct Lagrangian
- We impose as before feasibility and govt. budget constraint.
- Plus, optimal choices by **each** kind of consumer i .
- Plus, in model with many agents, that feasibility and govt. budget constraint hold does not imply that each individual budget constraint holds.
- Therefore, we put them as constraints.

- Lagrangian for Ramsey Problem:

$$\begin{aligned}
 \mathcal{L} = & \sum_{t=0}^{\infty} \beta^t \left\{ \sum_{i=1}^N \alpha_i u_i(c_{it}, 1 - n_{it}) + \right. \\
 & \psi_t [F(k_t, n_t) - \bar{r}_t k_t - \bar{w}_t n_t - g_t - S_t] + \\
 & \theta_t [F(k_t, n_t) - c_t - k_{t+1} + (1 - \delta)k_t - g_t] + \\
 & + \sum_{i=1}^N \mu_{i1t} [u_{ilt} - u_{ict} \bar{w}_t] \\
 & + \sum_{i=1}^N \mu_{i2t} [u_{ict} - \beta u_{ict+1} (\bar{r}_{t+1} + 1 - \delta)] \\
 & \left. + \sum_{i=1}^N \varepsilon_{it} [\bar{w}_t n_{it} + \bar{r}_t k_{it} + S_{it} - c_{it} - k_{it+1} + (1 - \delta)k_{it}] \right\}
 \end{aligned}$$

- First-order condition with respect to k_{it+1} :

$$\begin{aligned}
 \theta_t + \varepsilon_{it} = & \beta \{ \psi_{t+1} [F_k(t+1) - \bar{r}_{t+1}] + \theta_{t+1} [F_k(t+1) + 1 - \delta] \\
 & + \varepsilon_{it+1} (\bar{r}_{t+1} + 1 - \delta) \}
 \end{aligned}$$

- Derivative has same structure as with one agent, plus two terms that come from individual budget constraint.
- Assume a steady-state equilibrium exists.
- Steady-state FOC for capital (substituting $F_k = r$):

$$\theta + \varepsilon_i[1 - \beta(\bar{r} + 1 - \delta)] = \beta \{ \psi[r - \bar{r}] + \theta[r + 1 - \delta] \}$$

- From individual Euler equation in steady state,
 $1 - \beta(\bar{r} + 1 - \delta) = 0.$
- Therefore, $r = \bar{r}$, which again implies $\tau_k = 0.$

Back to Judd's question:

- Consider an extreme case of heterogeneity with only two classes of agents.
- Class 1 are workers who do not save. Budget constraint:
$$c_{1t} = \bar{w}_t n_{1t} + S_{1t}$$
- Class 2 are capitalists who do not work. Budget constraint:
$$c_{2t} + k_{2t+1} - (1 - \delta)k_{2t} = \bar{r}_t k_{2t} + S_{2t}.$$
- Assume $\alpha_1 > \alpha_2 = 0$; Ramsey cares only about workers.
- This economy is a special case of the general framework.
- Therefore result is the same: do **not** tax capital (owners).
- Government spending financed with tax on labor.
- Result breaks down if discount factors different for each agent ($\beta_i \neq \beta_j$), as Euler equation would hold for only 1 agent.

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